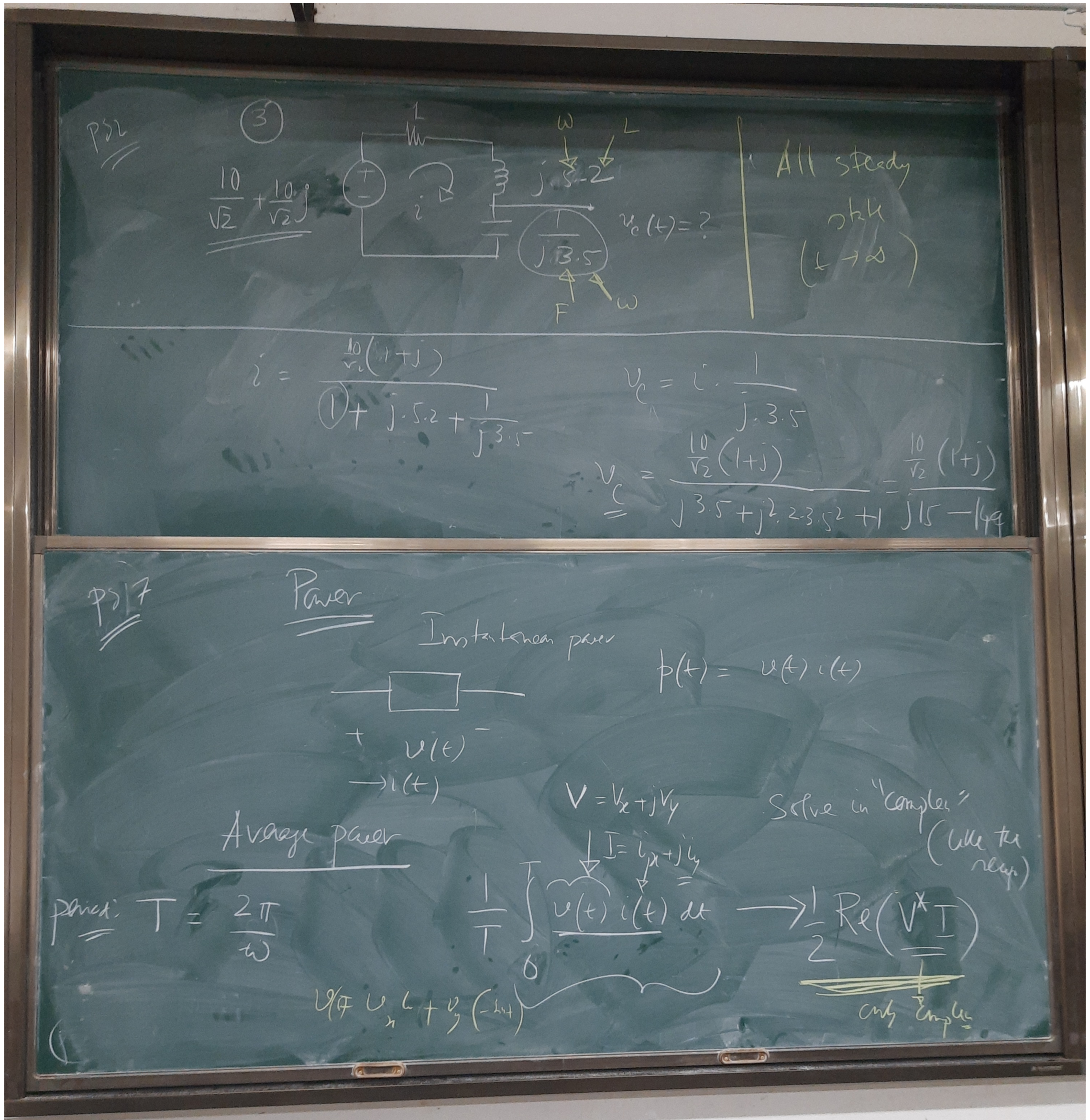


PS2

③

$\frac{10}{\sqrt{2}}+\frac{10}{\sqrt{2}}j$

$j\cdot5\cdot2$

$\frac{1}{j\cdot3\cdot5}$

$v_c(t)=?$

All steady state ($t\to\infty$)

$$i=\frac{\frac{10}{\sqrt{2}}(1+j)}{1+j\cdot5\cdot2+\frac{1}{j\cdot3\cdot5}}$$

$$v_c=i\cdot\frac{1}{j\cdot3\cdot5}$$

$$v_c=\frac{\frac{10}{\sqrt{2}}(1+j)}{j3\cdot5+j^2\cdot2\cdot3\cdot5^2+1}=\frac{\frac{10}{\sqrt{2}}(1+j)}{j15-149}$$

PS 7

Power

Instantaneous power

$+\ v(t)\ -$

$\rightarrow i(t)$

$p(t)=v(t)\,i(t)$

Average power

period: $T=\frac{2\pi}{\omega}$

$V=V_x+jV_y$

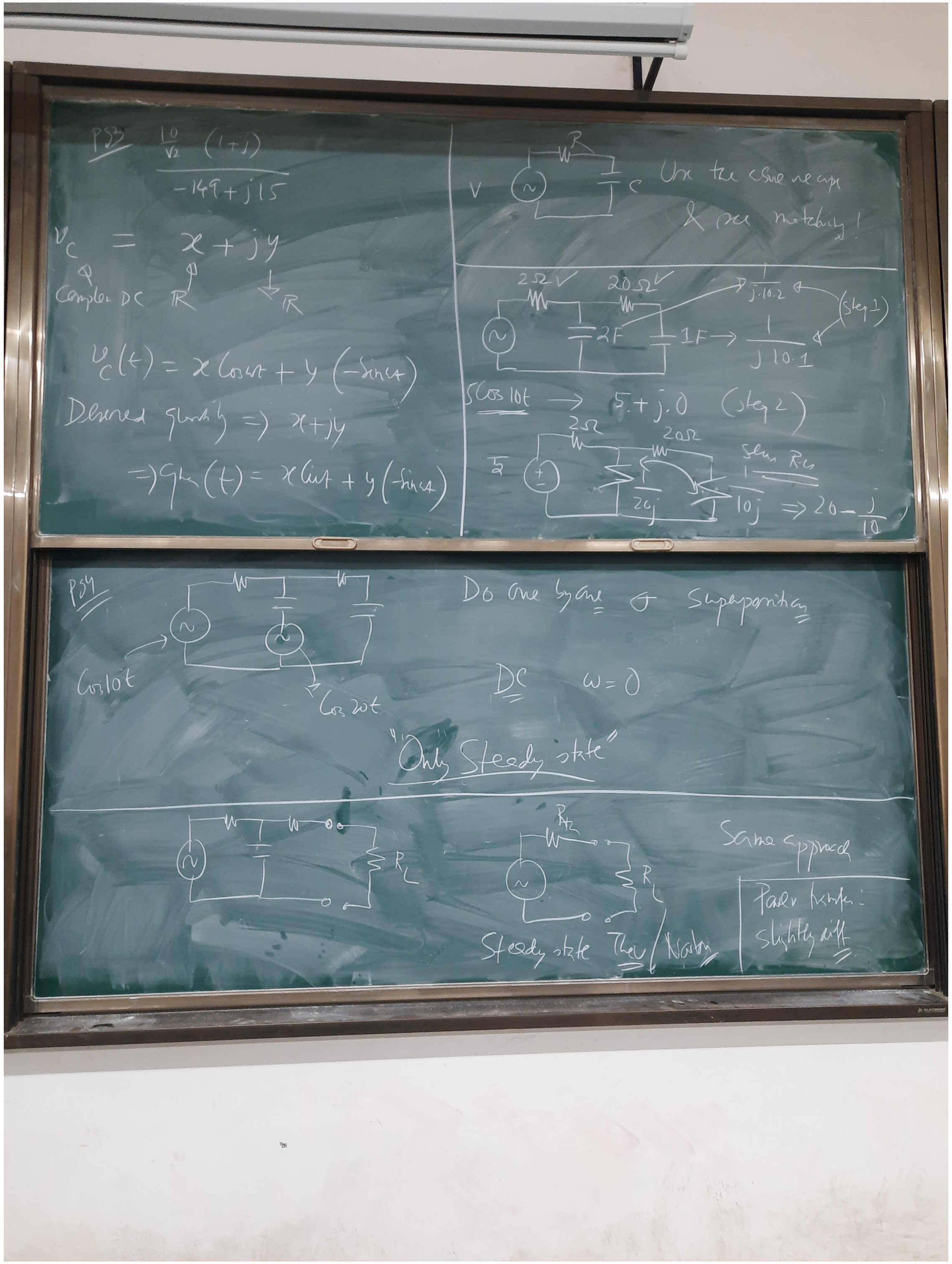
$I=i_x+ji_y$

Solve in "complex" (like the recap)

$$\frac{1}{T}\int_0^T \underbrace{v(t)\,i(t)}\,dt \longrightarrow \frac{1}{2}\mathrm{Re}(V^{*}I)$$

$v(t)$ $v_x\cos + v_y(-\sin t)$

only complex



PS3

$$\frac{\frac{10}{\sqrt{2}}(1+j)}{-149+j15}$$

$v_C = x + jy$

Complex DC ℝ ℝ

$v_C(t) = x\cos\omega t + y(-\sin\omega t)$

Desired quantity ⇒ $x+jy$

⇒ $q_{th}(t) = x\cos\omega t + y(-\sin\omega t)$

Use the same recipe

& see matching!

$1F \rightarrow \frac{1}{j\cdot 10\cdot 1}$

$\frac{1}{j\cdot 10\cdot 2}$

(step 1)

$5\cos 10t \rightarrow 5 + j\cdot 0$ (step 2)

Series R-C

$\frac{1}{20j}$

$\frac{1}{10j} \Rightarrow 20 - \frac{j}{10}$

PS4

Do one by one ⇒ Superposition

DC $\omega = 0$

"Only Steady state"

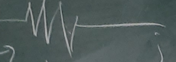
Steady state Thev / Norton

Same approach

Power transfer: slightly diff

PSS

$I \rightarrow$

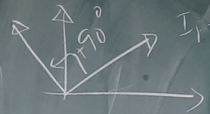


$\frac{1}{j\omega C} = -\frac{j}{\omega C}$

$V = I \cdot \left(\boxed{-j}\ \frac{}{\omega C}\right)$

Inductor

$j\omega L$



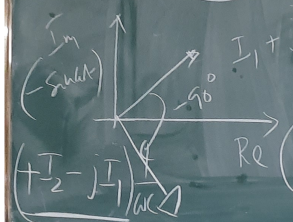
What it means in real?

(Trigonometric)

$I = I_1 \cos\omega t + I_2(-\sin\omega t)$

$= \sqrt{I_1^2 + I_2^2}\ \cos\left(\omega t + \tan^{-1}\frac{I_2}{I_1}\right)$

$I_1 + jI_2$



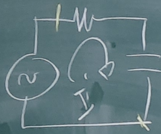
$V = \frac{1}{\omega C}\left(I_2\cos\omega t - I_1(-\sin\omega t)\right)$ predict

$= \frac{1}{\omega C}\sqrt{I_1^2 + I_2^2}\ \cos\left(\omega t + \tan^{-1}\left(\frac{I_2}{I_1}\right)\boxed{-90^\circ}\right)$ phase lag

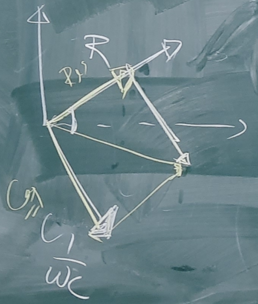
Inductor Voltage leads by 90°

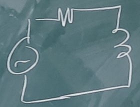
Cap voltage lags by 90°

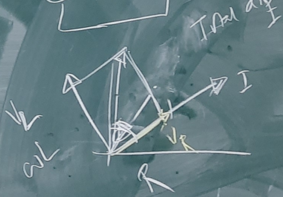
} Irr. of freq (→0)

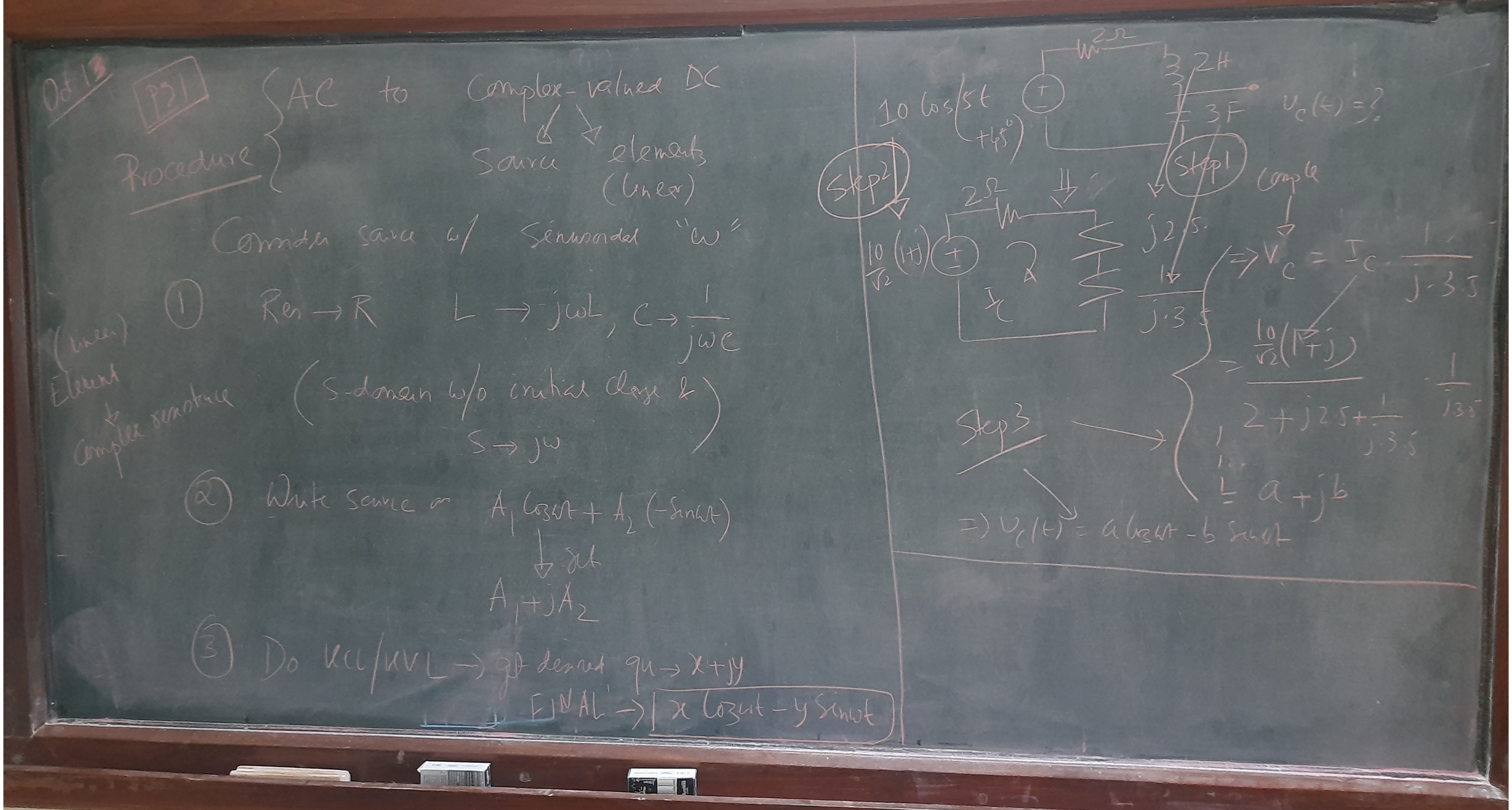


Phasor diagram









Oct 13

PS1

Procedure { AC to Complex-valued DC → Source, elements (linear)

Consider source w/ sinusoidal "ω"

① Res → R $L \to j\omega L$, $C \to \frac{1}{j\omega C}$

(linear) Element → complex resistance

(s-domain w/o initial charge & $s \to j\omega$)

② Write source as $A_1 \cos\omega t + A_2(-\sin\omega t)$

$$\xrightarrow{\text{set}} A_1 + jA_2$$

③ Do KCL/KVL → get desired qu → $x + jy$

FINAL → $\boxed{x \cos\omega t - y \sin\omega t}$

Step 3 →

$$\Rightarrow V_c = I_c \cdot \frac{1}{j\cdot3\cdot5}$$

$$= \frac{\frac{10}{\sqrt{2}}(1+j)}{2 + j2\cdot5 + \frac{1}{j\cdot3\cdot5}} \cdot \frac{1}{j\cdot3\cdot5}$$

$$\doteq a + jb$$

$\Rightarrow v_c(t) = a \cos\omega t - b \sin\omega t$